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Reaching the public with Twitter: The reputation value of CEOs

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ABSTRACT

In this study, we investigate whether CEOs' use of Twitter influences their compensation. The results show that CEOs benefit from being active on social media because this participation helps reduce information asymmetry. Further, the effect of this Twitter usage becomes stronger when firms face competitive markets. We also use earnings announcements to show how CEOs' tweets can lower the deviations in stock prices around these announcements. Overall, a CEO's personal use of social media can help to reduce the information asymmetry between their firm and investors. This strategic tool benefits the CEO when negotiating compensation.

1. Introduction

A CEO's personal image plays an important role in an organization's success because that image is the most valuable and competitive human resource. In order to maintain a good image with the public, CEOs have used inefficient media tools like the coverage by the news media in the past (Deephouse, 2000; Francis, Huang, & Zang, 2008). Nowadays, CEOs can use personal social media to establish their reputations and enhance their relationships with stakeholders, which can lead to positive outcomes (Weber, 2014). These personal uses of social media differ from the traditional media setting because CEOs may regulate the frequency of public appearances, so they can regularly communicate with investors rather than depending on the news cycle (Kelton & Pennington, 2020). However, we still have a limited understanding of how the use of social media can really benefit CEOs.

In this study, we examine the effect of CEOs' personal use of social media on their compensation by focusing on the use of Twitter. Recently, Twitter has become crucial to firms as a new source of information that can improve their public relations (Bushee & Miller, 2012; Miller, 2006; Miller & Skinner, 2015). Compared with traditional media, Twitter provides timely and value-relevant information to improve a firm's information environment (Bartov, Faurel, & Mohanram, 2018; Chen, Hwang, & Liu, 2017, 2018; Elliott, Grant, & Hodge, 2018).

By using Twitter, a firm is able to manage its information asymmetry in a timely manner. Blankespoor, Miller, and White (2014)

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focused on information technology firms with active Twitter accounts and found that the dissemination of information via tweets was associated with lower bid-ask spreads, greater depths, and a higher liquidity ratio. They showed that firms used Twitter to spread news, and this dissemination helped to reduce investors' information acquisition costs and allowed more potential investors to process the information. Similarly, corporate executives are increasingly active on Twitter to directly communicate with investors. Chen, Hwang, and Liu (2018) suggested that tweets coming from top executives were more impactful than those coming from firms.

To understand the effect of CEOs' Twitter use on their compensation, we hand-collect top executives with personal Twitter accounts from the Execucomp database for the period from 2006 to 2020.[1] Through this labor-intensive process, we identify 336 CEOs (13% of all CEOs) that have active personal Twitter accounts and sufficient information on the accounts of their firms to help address omitted variable issues.[2] We obtained all accounting variables and stock prices from the Compustat database and the Center for Research in Security Prices (CRSP). We also collect corporate-governance and CEO-compensation-related variables from RiskMetrics and the Execucomp database.

We document three sets of findings from these data. First, we find that CEOs with personal Twitter accounts have higher total compensation. That is, these CEOs are likely to have higher compensation after controlling for CEO characteristics, firm characteristics, and corporate governance factors. The evidence remains strong when we control for industry and year fixed effects. We hereafter abbreviate the effects of the CEO's use of a personal Twitter account on firm value as *CEO Twitter usage* to ease the discussion.

We further examine the underlying channels between Twitter usage and the compensation of CEOs. We first use several variables such as firm age, number of analysts covering, and rating to identify the cause of the information asymmetry. We argue that a personal Twitter account can be viewed as an efficient tool for spreading information, and firms with higher information asymmetry are more likely to use it. Our results support this argument by confirming that CEO Twitter usage happens primarily in firms with high information asymmetry. We show that the usage has a stronger effect if the firm is younger and has lower analyst coverage. Hence, our results show that the effects of CEO Twitter usage on compensation can derive from the information asymmetry between executives and shareholders.

Another possible cause is market competition. Compared with studies that focus on how market competition affects information asymmetry, we investigate whether CEO Twitter usage has a different effect on firms experiencing greater threats in the product market (Akins, Ng, & Verdi, 2012; Easley & O'hara, 2004; Kyle, 1985). We separate firms into high and low competition groups by using several proxies for the market competition, such as consumer product-oriented, product similarity, and product market fluidity.[3] The results show that the positive relationship between Twitter use and compensation occurs only for the group with high market competition. In sum, information asymmetry and market competition can help explain why CEO Twitter usage can raise their compensation.

We next analyze the role of CEO Twitter usage by focusing on the market response to earning announcements. We regress the cumulative abnormal return (*CAR*) around earnings announcements on standardized unexpected earnings (*SUE*), which is a proxy for information asymmetry. We further interact *SUE* with CEO Twitter usage. The results show that the interaction is negatively related to the *CAR* that indicates the CEOs' personal tweets can curtail the deviation in stock prices around earnings announcements.

We use several additional tests to examine the robustness of our main results. First, we use an instrumental variable in the two-stage regression model to address the endogeneity from a CEO's personal Twitter account suffering from selection bias. Second, we use a placebo test by replacing our key variable of interest with lagged variables to see whether CEO Twitter usage happens only after CEOs create Twitter accounts. Third, we consider alternative measures of compensation (Delta, Vega, firm related wealth, and non-equity-based compensation) and alternative empirical design. Finally, we focus on the effects of Twitter usage on revenue efficiency and managerial ability. To conclude, our main findings continue to hold when using these alternative measures.

Our study contributes to the literature from several perspectives. First, it is related to the studies on social media. Our findings highlight the benefit of using Twitter as a direct communication channel for information. This finding makes two important contributions. First, it has important implications for the effect that Twitter usage has on executives' compensation. Most studies that address the issues related to CEO compensation are based on the agency theory; our results indicate that CEOs' use of Twitter can create an important social network that mutually benefits them and investors.

Second, our study contributes to a growing group that focuses on the importance of CEO characteristics to corporate decisions (Malmendier & Tate, 2005; 2008; Malmendier & TateGYan, 2011; Hirshleifer, Low, & Teoh, 2012). The information from a CEO's tweets can help investors make their investment decisions. Thus, Twitter can play a role in making the market more efficient by uncovering value-relevant information. This role is stronger for firms in weak information environments in which regulatory intervention seems not to help.

Third, we contribute to the literature on information asymmetry. Studies have examined this issue in different topics such as disclosure (Brown & Hillegeist, 2007; Shroff, Sun, White, & Zhang, 2013), media coverage (Fang & Peress, 2009), cost of capital (Armstrong, Core, Taylor, & Verrecchia, 2011), and financial reporting frequency (Fu, Kraft, & Zhang, 2012). Our study adds to this growing literature by linking information asymmetry to CEOs' social media usage and compensation.

The rest of the paper is organized as follows: In Section 2, we develop the hypotheses; Section 3 presents the data sources, main variables, and the method; Section 4 offers empirical results; Section 5 provides several robustness checks; and Section 6 concludes the

[1] Execucomp covers the S&P 1500 as well as companies that were once part of the S&P 1500 index.

[2] We collected the Twitter data, which identifies how many CEOs had used Twitter, through Twitter API in September 2018. We are unable to expand this Twitter data afterward because Twitter stopped sharing its API data in February 2023.

[3] See Hoberg and Phillips (2016), Hoberg et al. (2014), and Hoberg and Phillips (2010).

paper.

2. Literature review and hypothesis development

Before Twitter, it was hard to find a proper platform that could efficiently disseminate a CEO's personality to investors (Gallaugher & Ransbotham, 2010). These platforms included Google search (Da, Engelberg, & Gao, 2011; Drake, Roulstone, & Thornock, 2012); Internet bulletin boards (Hirschey, Richardson, & Scholz, 2000; Tumarkin & Whitelaw, 2001); and message boards, such as Yahoo! or Raging Bull, (Antweiler & Frank, 2004; Das & Chen, 2007). Later, social media adjusted this drawback (e.g., Facebook, StockTwits, Twitters) by adding a user's personal information that helped to identify their personality.

The change in social media further linked a CEO's inherent personality to real outcomes in the financial markets (Bollen, Mao, & Zeng, 2011; Chen, De, Hu, & Hwang, 2014; Jung, Naughton, Tahoun, & Wang, 2018; Laroche, Habibi, & Richard, 2013; Lee, Hutton, & Shu, 2015; Tetlock, 2007). For example, Curtis, Richardson, and Schmardebeck (2016) showed that a CEO's greater engagement with social media would cause a greater sensitivity of earnings announcement returns to earnings surprises, while a lower level of social media engagement was associated with considerable post-earnings-announcement drift. Additionally, better social media engagement could benefit a firm by raising its long-run productivity and equity valuation (Luo, Zhang, & Duan, 2013). In sum, firms increasingly have used social media to reduce information asymmetry, improve securities' liquidity, and lower bid-ask spreads (Blankespoor et al., 2014; Bushee & Leuz, 2005; Diamond & Verrecchia, 1991; Leuz & Verrecchia, 2000; Verrecchia, 2001; Xu & Zhang, 2013).

Among these social media, Twitter undoubtedly is one of the most popular platforms in the US. Jung et al. (2018) collected data from all firms included in the S&P 1500 index and showed that Twitter had become the preferred social media platform for firms. By requiring a short format (140-character limitation), Twitter eases the process of releasing information by giving users the opportunity to communicate and share opinions in a timely fashion. The text content of Twitter, therefore, may include day-to-day activities, current interests, or a personal mood (Naaman, Boase, & Lai, 2010).

Studies have indicated how a firm might benefit from the use of Twitter. By using text processing techniques, Bollen et al. (2011) found that the collective mood stated on Twitter could predict changes in the value of the Dow Jones Industrial Average over time. Consistently, Mao, Wei, Wang, and Liu (2012) have shown that the daily number of tweets that mention S&P 500 stocks was significantly associated with the S&P 500's daily closing prices, its daily price changes, and its daily absolute price changes. Blankespoor et al. (2014) also indicated that firm-initiated news via Twitter was associated with lower bid-ask spreads, greater depths, and higher stock liquidity.

Besides the firm's own social media pages, a CEO may also benefit from the use of social media. Compared with the official press, CEO can write their own stories and cultivate their own followers on Twitter. These social activities help CEO build their public image through their timely performance and fostering of celebrity culture. As a result, a CEO has the incentive to use social media such as Twitter. These efforts also benefit firms. Karaduman (2013) showed that the personal branding efforts of top-level executives could create value not only for themselves but also for their firms. Chen et al. (2018) indicated that tweets coming out of top executives' personal Twitter accounts were more effective than those coming from the firms' managed accounts.

To conclude, these efforts on social media should lead to better compensation for CEOs based on the positive relationship between CEO awards and compensation (Malmendier & Tate, 2009). This hypothesis is identical to other motives related to agency or managerial perspectives because CEOs are not taking advantage of shareholders (Banker, Darrough, Huang, & Plehn-Dujowich, 2013; Bugeja, Matolcsy, & Spiropoulos, 2012; Chhaochharia & Grinstein, 2009; Hartzell & Starks, 2003; John, Mehran, & Qian, 2010). Therefore, we propose our first hypothesis:

H1. CEOs who use Twitter are likely to get higher compensation (the effect of CEO Twitter usage).

Because the use of Twitter may affect firms' value mainly by reducing information asymmetry, the role of information disclosure matters when discussing the effects of Twitter (Blankespoor et al., 2014; Tetlock, 2010).[4] We can find similar evidence in capital markets. For example, Armstrong et al. (2011) argued that information asymmetry played an important role in the cost of capital when the market competition was low. Tang (2009) found that the release of credit ratings allowed lenders to evaluate borrowers' potential risks that reduced information asymmetry. Baik, Kang, and Kim (2010) analyzed the role of geographic proximity and found that institutional investors had a greater level of home bias for firms with higher information asymmetry. Boone and White (2015) also showed that the following analysts could help institutional investors enhance monitoring efficiency by reducing information asymmetry and trading costs.

According to these discussions, the use of Twitter is likely to enhance firm value by reducing the costs of capital when information asymmetry is high. Because the CEO's image affects not only their own reputation but also that of firms, we predict that firms are more likely to recruit CEOs with better images on social media when facing a higher level of information asymmetry. Thus, we propose our next hypothesis:

[4] See the studies of Welker (1995) and Healy and Palepu (2001) on the disclosure policy and liquidity; Lang et al. (2000), Brown and Hillegeist (2007), and Shroff et al. (2013) for the literature on disclosure; Chae (2005) on that trading volume and corporate announcements; Hilary (2006) on the organized labor and management; Moeller, Schlingemann, and Stulz (2007) for evidence on acquirer returns; Fang and Peress (2009) on media coverage and stock returns; Tetlock (2010) for evidence on public financial news; Armstrong et al. (2011) for the cost of capital; Fu et al. (2012) on the financial reporting frequency and cost of equity.

H2. The effect of CEO Twitter usage is stronger for firms with higher information asymmetry.

Information can also be a strategic tool when competing with other firms. This strategy is important because market competition can largely alter a firm's decisions when there is no effective channel to lower the pressure of competition. Hoberg, Phillips, and Prabhala (2014) argued that competition might threaten firms and make them repurchase fewer shares, pay lower dividends, and hold more cash and liquid assets. Hoberg and Phillips (2010) also showed that firms might change the frequency of their product launch to maintain their profits when facing threats from rivals.

To understand the relationship between information asymmetry and market competition, Akins et al. (2012), Easley and O'hara (2004), and Kyle (1985) showed that the increase in market competition could reduce information asymmetry because competition makes information spread more quickly to uninformed investors. Such effects also influence the process of finding the equilibrium price. Armstrong et al. (2011) also found that information asymmetry could affect the costs of capital when a firm faced imperfect market competition because competition helped dig out private information.

Compared with studies that identify the effects of market competition on information asymmetry, we predict that the reduction in information asymmetry can give a firm an advantage when competing with other firms. Additionally, a firm can reduce information asymmetry by managerially releasing positive information to alter its image. These strategic reasons make the use of social media more important when facing market competition. We propose our third hypothesis as follows:

H3. The effect of CEO Twitter usage is stronger for firms with higher market competition.

When clarifying the effect of Twitter usage on firm value, we used earnings announcements as targets because they are the most frequent announcements by a firm. The deviation of abnormal returns around the earnings announcement also creates a significant change in a firm's value (May, 1971; Kane, Lee, & Marcus, 1984; La Porta, Lakonishok, Shleifer, & Vishny, 1997; Fama, 1998; Kaniel, Liu, Saar, & Titman, 2012). For example, La Porta et al. (1997) showed that earnings announcements can create, on average, a 20% change in stock returns.

When the use of Twitter has a negative relationship with information asymmetry, we predict that evidence will exist around an earnings announcement. For example, Bartov et al. (2018) argued that Twitter could be a useful tool for predicting abnormal returns around earnings announcements. The role of tweets in predicting announcements is more prominent in a high information asymmetry environment. We can find similar evidence for this argument when focusing on other types of announcements. Lee et al. (2015) showed that the use of Twitter by a firm could weaken the negative price reaction to recall announcements by broadcasting the firm's message directly to investors without distorting the content. When managing the information release properly, firms can also engage users and customers through social media to mitigate the adverse effect of the bad reputation associated with product recalls.

When a firm realizes that Twitter can be an efficient tool to reduce its information asymmetry, we predict that it may use Twitter to release information around earnings announcements. This information also helps curtail the market responses to the announcements. We propose our fourth hypothesis as follows:

H4. The effect of CEO Twitter usage is more likely to curtail the market responses to earnings announcements.

3. Data and summary statistics

3.1. Data and variables

To construct our sample of CEOs with personal Twitter accounts, we download a list of all CEOs from the Execucomp database for the period from 2006 to 2020. Execucomp covers the S&P 1500 as well as firms that were once part of the S&P 1500 index but are still trading. We start with the list of all CEOs in Execucomp and identify those with active Twitter accounts by matching the first and last names on the account with the list. We then cross-check the executives' middle names, gender, and firm information with user characteristics; we also read tweets to check whether these accounts release their firms' information.

Through this labor-intensive process, we find 336 CEOs who have active personal Twitter accounts and work for firms that have the data necessary to conduct our tests. Our main explanatory variable is an indicator *CEO_Twitter* that identifies whether the executive has a Twitter account. It equals one if the CEO of firm i adopted Twitter after year t, and zero otherwise. We exerted great effort in separating out Twitter accounts managed by firms. Our sample comprises 16,684 observations from 2006 to 2020.

The main dependent variable is the level of annual CEO compensation collected from the Execucomp database. CEO compensation is measured using three different variables: total annual compensation, annual salary, and annual bonus. Total compensation (Execucomp data item: TDC) comprises a CEO's salary, bonus, other annual compensation, total value of restricted stock granted, total value of stock options granted (valued using Black–Scholes), long-term incentive payouts, and all other compensation (in thousands of dollars). We use the natural logarithm of Execucomp's total compensation (*TDC*) as the main measure to ease the interpretation. Appendix A provides the definitions of all variables.

We also identify other important characteristics according to Bugeja et al. (2012) and Chhaochharia and Grinstein (2009). These variables controlled for CEO perspectives, firm characteristics, and corporate governance that were associated with CEO compensation. CEOs' perspectives comprise the natural logarithm of their tenure (*CEO_tenure*); an indicator that equals one if the CEO is also the board chairperson (*CEO_chair*), and zero otherwise; an indicator that equals one if it is the CEO's first year at a firm (*CEO_firstyear*), and zero otherwise; an indicator that equals one if the CEO owns 5% or more of the firm's stock (*CEO5pct*), and zero otherwise.

Firm characteristics are from Compustat and comprise the book-to-market ratio (*BMV*), annual common stock return (*RET*), return on assets (*ROA*), natural logarithm of the standard deviation of common stock returns (*Std5RE)*), natural logarithm of the standard

deviation of the return on assets measured over the previous five years (*Std5ROA*), and the liabilities divided by equity (*DE*). Corporate governance is from Risk Metrics and comprises the number of board directors (*Bdsize*); an indicator that equals one if the compensation committee comprised only independent directors (*Indcompcom*), and zero otherwise. We further control for industry and year fixed effects and adjust standard errors with within-firm clustering (White, 1980).

3.2. Summary statistics

Table 1 presents the summary statistics of all variables. The mean value of *CEO_Twitter* is 0.136, with a standard deviation of 0.343. This value shows that only 13.6% of CEOs use Twitter. In terms of CEO perspectives, the mean of *TDC* is 8.415 (around USD 3 million) with a standard deviation of 1.123. The average number of years the CEO held their position at a firm is 2.182 (around 8.9 years) in the sample.[5]

We further compare the Twitter and non-Twitter users in Table 2 to understand whether other CEO perspectives drive the difference between them. We also compare the other main variables to see if their omission causes the difference. In the group with a Twitter account, the mean of *TDC* is 8.550, which is higher than that in the group of non-Twitter users by 0.157. Therefore, Twitter users, on average, receive higher compensation than those who are not. We do not find that Twitter users have different working experiences from other CEOs, but they are less likely to rule the board as indicated by *CEO_chair*. The results show that Twitter users are not likely to receive higher compensation because they have greater power over the compensation committee.

In terms of firm characteristics, the CEOs with Twitter accounts have greater operating (*ROA*) and short-run performance (*RET*). However, these firms have less growth opportunity (*BMV*). A firm's growth opportunities are not likely to explain the high compensation of Twitter users either.

We also report Pearson's correlation coefficient matrix in Table 3. As expected, we find a positive and significant correlation between *TDC* and *CEO_Twitter*. The correlation coefficients between *TDC* and most control variables are significant and show that these variables can identify other driving factors behind a CEO's compensation well.

4. Empirical results

4.1. Baseline result: Twitter use and compensation of CEOs

According to Chhaochharia and Grinstein (2009), we used the following regression model to test H1:

$$TDC_{i,t} = \alpha_0 + \alpha_1 CEO_Twitter_{i,t} + \beta' F_{i,t-1} + \theta' Z_{i,t} + \gamma_i + \mu_t + \varepsilon_{i,t}, \quad (1)$$

where the dependent variable is *TDC* and the main explanatory variable is *CEO_Twitter*. Specifically, $TDC_{i,t}$ is the natural logarithm of compensation for CEO i in year t; $CEO_Twitter_{i,t}$ is an indicator that equals one when a CEO has a Twitter account, and zero otherwise. Also, this measure can capture the difference in compensation for Twitter users before and after the use of Twitter. $F_{i,t-1}$ is a vector of firm characteristics for firm i in year t-1. $Z_{i,t}$ is the vector of control variables for CEO perspectives and corporate governance. The γ_i and μ_t represent the fixed effects of industry and year, respectively, and $\varepsilon_{i,t}$ represents the error term. In all specifications, the *t-statistics* are based on heteroscedasticity among the firm-level, robust standard errors (White, 1980).

Table 4 presents the results. In order to address any omitted variables in the results, we conduct four specifications with different settings of fixed effects and control variables. We find that *CEO_Twitter* positively and significantly affects *TDC* after controlling for all potential factors, indicating that CEOs who use Twitter get higher compensation. Thus, the results support H1 that *CEOs who use Twitter are likely to get higher compensation*. According to the results in column (4), CEOs who use Twitter have 0.140 more *TDC* than those who do not, which is around one standard deviation in *TDC*.

The results of the control variables are generally consistent with the literature (Bugeja et al., 2012; Chalmers, Koh, & Stapledon, 2006; Core, Holthausen, & Larcker, 1999). Compensation is negatively associated with *CEO_firstyear*, which is consistent with the idea that CEOs receive significantly lower total compensation in the first year. As shown by *CEO5pct*, CEOs who own more than 5% of stock are paid significantly less. This finding can be explained by agency costs: CEOs with lower ownership are likely to extract higher compensation because they have less incentive to maximize their firm's market value. Compensation is positively associated with *CEO_chair* because CEOs with greater power ask for higher compensation.

The results of the measures related to corporate governance are consistent with our prediction, too (Bugeja et al., 2012; Core et al., 1999; Wade, O'Reilly, & Chandratat, 1990). CEOs are likely to receive higher compensation when a firm has higher growth opportunities and better performance. The effects of control variables show a similar pattern as reported in the literature.

Overall, we find that CEOs who use Twitter get higher compensation compared with those who do not. Consistent with our prediction, CEOs are compensated when using their personal Twitter accounts to display their ability to convey information on social media. In addition to Karaduman's (2013) findings that CEO's branding efforts benefit their firms, we show that firms also feed back to their CEOs for such branding efforts. These feedbacks are non-trivial.

[5] Appendix B supplies the sample distribution by SIC classification.

Table 1
Descriptive Statistics
This table presents the descriptive statistics of all variables used in this research. The sample period is from 2006 to 2020. *TDC* is the total compensation and consists of salary, bonus, the value of restricted stock granted, the value of options granted (using Black– Scholes), long-term incentive payouts, and other compensation. *CEO_Twitter* is an indicator for CEOs who use Twitter that equals one when CEOs have Twitter accounts, and zero otherwise. *CEO5pct* is an indicator variable that equals one if the CEO owns 5% or more of the firm's stock, and zero otherwise. *CEO_firstyear* is an indicator variable that equals one if it is the CEO's first year of service at a firm, and zero otherwise. *CEO_tenure* is the natural logarithm of their tenure. *CEO_chair* is an indicator variable that equals one when the CEO is also the board chairperson, and zero otherwise. Other variables are defined in Appendix A.

Variables	Mean	p25	p50	p75	Sd
TDC	8.415	7.858	8.509	9.092	1.123
CEO_Twitter	0.136	0.000	0.000	0.000	0.343
CEO5pct	0.866	1.000	1.000	1.000	0.341
CEO_firstyear	0.051	0.000	0.000	0.000	0.220
CEO_tenure	2.182	1.946	2.303	2.565	0.531
CEO_chair	0.538	0.000	1.000	1.000	0.499
Bdsize	9.421	8.000	9.000	11.000	2.287
Indcompcom	0.962	1.000	1.000	1.000	0.192
BMV	0.660	0.454	0.659	0.875	0.265
DE	0.921	0.162	0.524	1.062	21.431
ROA	0.089	0.040	0.078	0.126	0.089
RET	0.104	−0.126	0.074	0.277	0.446
Std5RET	0.523	0.184	0.279	0.424	11.391
Std5ROA	0.031	0.009	0.019	0.036	0.047

Table 2
CEOs with and without Twitter account.
This table compares the differences in CEO, governance, and firm characteristics between CEOs with and without Twitter accounts. *CEO_Twitter* is an indicator for CEOs who use Twitter that equals one when CEOs have Twitter accounts, and zero otherwise. Other variables are defined in Appendix A. The *t*-test is deployed to compare the means. The *, **, and*** denote significance at 10%, 5%, and 1% levels, respectively.

Variables	With Twitter		Without Twitter		Differences
	Obs.	Mean (1)	Obs.	Mean (2)	(1)–(2)
TDC	2,270	8.550	14,414	8.393	0.157***
CEO5pct	2,270	0.842	14,414	0.869	−0.027***
CEO_firstyear	2,270	0.0480	14,414	0.0520	−0.004
CEO_tenure	2,270	2.187	14,414	2.181	0.006
CEO_chair	2,270	0.516	14,414	0.542	−0.026**
Bdsize	2,270	9.523	14,414	9.404	0.120**
Indcompcom	2,270	0.960	14,414	0.962	−0.002
BMV	2,270	0.596	14,414	0.670	−0.074***
DE	2,270	1.448	14,414	0.839	0.609
ROA	2,270	0.0960	14,414	0.0880	0.008***
RET	2,270	0.121	14,414	0.102	0.019*
Std5RET	2,270	0.0310	14,414	0.0310	−0.000
Std5ROA	2,270	0.416	14,414	0.540	−0.124

4.2. CEO Twitter usage and information asymmetry

We next examine the underlying channel of CEO Twitter usage. The empirical studies have indicated that information asymmetry causes a gap in the information between managers and shareholders. Following Chae (2005), Autore and Kovacs (2010), and Leary and Roberts (2010), we use several proxies, such as firm age (*Age*) and analyst coverage (*Coverage*), to measure the information asymmetry. We separate the whole sample into groups with high and low information asymmetries to examine whether CEO Twitter usage is more prominent in the group with high information asymmetry. The criteria for high information asymmetry are as follows: (1) firms whose ages are lower than the median and (2) firms with analyst coverage that is smaller than the median.

Table 5 presents the results after controlling for all control variables and fixed effects.[6] The result shows that *CEO_Twitter* is positively and significantly related to *TDC* only for the high information asymmetry group when using firm age and analyst coverage to identify information asymmetry, while the coefficient is insignificant in the low information asymmetry group. Moreover, we use the Chow test to confirm the validity of the findings regarding the high and low information asymmetry groups. This test shows a

[6] The results for columns (1) and (4) cover the sample period from only 2006 to 2016 because of the limitation for I/B/E/S database from the authors' authorizations. We also use the sample period from 2006 to 2016 when focusing on earnings announcements in Table 7 for the same reason.

Table 3
Correlation matrix
This table presents the Pearson correlation matrix between all variables. Appendix A provides the definitions of variables. The * and ** denote significance levels at 5% and 1%, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
(1)TDC	1.00													
(2)CEO_Twitter	0.05***	1.000												
(3)CEO5pct	−0.10***	−0.03***	1.00											
(4)CEO_firstyear	0.01	−0.01	−0.15***	1.00										
(5)CEO_tenure	0.02***	0.01	0.19***	−0.25***	1.00									
(6)CEO_chair	0.08***	−0.02**	0.03***	−0.10***	0.25***	1.00								
(7)Bdsize	0.27***	0.02**	−0.16***	0.04***	−0.02***	0.08***	1.00							
(8)Indcompcom	0.06***	−0.01	−0.01	0.02**	−0.01	0.01	0.01*	1.00						
(9)BMV	−0.09***	−0.10***	0.04***	0.01	−0.05***	−0.02**	0.15***	0.01	1.00					
(10)DE	−0.01	0.01	−0.01	0.01	−0.01	−0.01	0.01	0.01	0.02***	1.00				
(11)ROA	0.08***	0.03***	−0.05***	−0.01	0.02**	0.04***	−0.04***	−0.01	−0.56***	−0.02***	1.00			
(12)RET	0.04***	0.02*	0.09***	−0.01	0.05***	0.00	−0.04***	0.01	0.05***	−0.01	−0.01	1.00		
(13)Std5RET	−0.06***	−0.01	0.04***	−0.01	−0.04***	−0.08***	−0.19***	0.01	−0.13***	−0.01	−0.04***	0.05***	1.00	
(14)Std5ROA	−0.01	−0.01	0.01	−0.01	0.01	0.01	−0.01	0.01	0.02**	0.01	−0.01	−0.01	0.05***	1.00

Table 4
CEO Twitter usage and compensation
These tables present the results of the ordinary least squares (OLS) of CEO Twitter usage on their compensation.
$TDC_{i,t} = \alpha_0 + \alpha_1 CEO_Twitter_{i,t} + \beta' F_{i,t-1} + \theta' Z_{i,t} + \gamma_i + \mu_t + \varepsilon_{i,t}$
where the dependent variable is *TDC* (total compensation) and the main explanatory variable is *CEO_Twitter*. Specifically, $TDC_{i,t}$ is the natural logarithm of compensation for CEO *i* in year *t*; *CEO_Twitter* is an indicator for CEOs who use Twitter that equals one when CEOs have Twitter accounts, and zero otherwise. $F_{i,t-1}$ is a vector of control variables (firm characteristics) for firm *i* in year *t*-1. $Z_{i,t}$ is the vector of the control variables for corporate governance and CEO characteristics *i* in year *t*. γ_i and μ_t represent the fixed effects of industry and year, respectively, and $\varepsilon_{i,t}$ represents the error term of the regression. Other variables are defined in Appendix A. In all specifications, the *t-statistics* reported are based on heteroscedasticity among the firm-level, robust standard errors (White, 1980). The sample period is 2006–2016. The *, **, and *** denote the significance levels of 10%, 5%, and 1%, respectively.

	(1)	(2)	(3)	(4)
	TDC	*TDC*	*TDC*	*TDC*
CEO_Twitter	0.140*	0.147**	0.143**	0.140**
	(1.896)	(1.992)	(2.014)	(1.978)
CEO5pct	−0.213***	−0.322***	−0.273***	−0.253***
	(-5.411)	(-7.951)	(-6.673)	(-6.280)
CEO_firstyear	−0.014	−0.049	−0.040	−0.044
	(-0.391)	(-1.425)	(-1.202)	(-1.322)
CEO_tenure	0.051	0.036	0.067**	0.043
	(1.587)	(1.128)	(2.298)	(1.503)
CEO_chair	0.132***	0.201***	0.198***	0.181***
	(3.447)	(5.264)	(5.488)	(5.041)
Bdsize	0.124***	0.118***	0.141***	0.138***
	(10.041)	(9.776)	(10.623)	(10.570)
Indcompcom	0.353***	0.333***	0.296***	0.300***
	(5.437)	(5.208)	(4.687)	(4.889)
BMV				−0.292***
				(-3.423)
DE				−0.000
				(-0.261)
ROA				0.428**
				(1.994)
RET				0.150***
				(8.023)
Std5RET				−1.510***
				(-4.316)
Std5ROA				0.000
				(1.000)
Constant	6.888***	6.627***	7.040***	7.232***
	(45.668)	(35.912)	(36.129)	(35.650)
Control for				
Industry FE	NO	NO	YES	YES
Year FE	NO	YES	YES	YES
Obs.	16,684	16,684	16,684	16,684
Adj R^2	0.0854	0.123	0.182	0.193

significant difference between the coefficients for *CEO_Twitter* for the two groups.

These results support H2 that *the effect of CEO Twitter usage is stronger for firms with higher information asymmetry*. When a firm's CEO uses social media that reduces information asymmetry between the firm and fund providers, the firm with higher information asymmetry is more likely to award these efforts by raising CEO's compensation. In contrast, firms with low information asymmetry do not award such efforts.

4.3. CEO Twitter usage and market competition

We further extend another underlying channel, market competition, for the effect of CEO Twitter *usage*. We use several proxies for market competition, such as product similarity (*Similarity*), product market fluidity (*Fluidity*), and consumer product-oriented (*CP oriented*) according to (Hoberg et al., 2014; Hoberg and Phillips, 2010, 2016). We separate firms into two groups with strong and weak market competition. A firm is classified as having strong market competition when its product similarity or product market fluidity is greater than the median. In contrast, a firm is classified as having strong market competition when its *CP oriented* is smaller than the median. A firm is classified as having weak market competition when it does not satisfy the corresponding criteria.

Table 6 presents the results. We find consistent results that the effect of CEO Twitter usage appears only in the group with high market competition whether we use either measure of market competition. We also examine the difference in coefficients between high and low product market competition by using the Chow test. The results show a significant difference between the two groups when focusing on *Similarity* and *CP oriented*.

In sum, the results support H3 that *the effect of CEO Twitter usage is stronger when a firm faces greater market competition*. A CEO's

Table 5
CEO Twitter usage and information asymmetry
This table presents the results of the OLS estimation of the influence of CEO Twitter usage on their compensation by considering the subsample of information asymmetry. We separate firms into high and low information asymmetry groups. The criteria for information asymmetry are as follows: (1) firm age (*Age*) is lower than the median and (2) analyst coverage (*Coverage)* is smaller than the median, and the low information asymmetry is vice versa. The equation is as follows:

$$TDC_{i,t} = \alpha_0 + \alpha_1 CEO_Twitter_{i,t} + \beta' F_{i,t-1} + \theta' Z_{i,t} + \gamma_i + \mu_t + \varepsilon_{i,t}$$

where the dependent variable is *TDC* (total compensation) and the main explanatory variable is *CEO_Twitter*. Specifically, $TDC_{i,t}$ is the natural logarithm of compensation for CEO *i* in year *t*; *CEO_Twitter* is an indicator for CEOs who use Twitter that equals one when CEOs have Twitter accounts, and zero otherwise. $F_{i,t-1}$ is a vector of control variables (firm characteristics) for firm i in year *t*-1. $Z_{i,t}$ is the vector of the control variables for corporate governance and CEO characteristics *i* in year *t*. The γ_i and μ_t represent the fixed effects of industry and year, respectively, and $\varepsilon_{i,t}$ represents the error term of the regression. In all specifications, the *t-statistics* reported are based on heteroscedasticity among the firm-level, robust standard errors (White, 1980). Appendix A provides the definitions of variables. The *, **, and *** denote the significance level of 10%, 5%, and 1%, respectively.

	High information asymmetry		*Low information asymmetry*	
	Age < Median	Coverage < Median	Age ≥ Median	Coverage ≥ Median
	(1)	(2)	(3)	(4)
CEO_Twitter	0.147*	0.1220***	0.127	−0.0607
	(1.692)	(3.42)	(1.043)	(-0.93)
CEO5pct	−0.192***	−0.3265***	−0.252***	0.1247**
	(-4.364)	(-7.92)	(-3.612)	(2.36)
CEO_firstyear	−0.103***	−0.1524**	0.038	0.003
	(-2.729)	(-2.48)	(0.687)	(0.03)
CEO_tenure	0.028	−0.0650***	0.080**	−0.0779**
	(0.706)	(-4.82)	(2.076)	(-2.96)
CEO_chair	0.180***	0.1337***	0.161***	0.1887**
	(4.118)	(3.67)	(2.899)	(2.95)
Bdsize	0.158***	0.1296***	0.106***	0.0680***
	(14.722)	(24.03)	(4.777)	(4.65)
Indcompcom	0.239***	0.2893***	0.321***	0.3973**
	(2.669)	(4.54)	(3.459)	(2.87)
BMV	−0.335***	−0.2362***	−0.343***	−0.2723***
	(-2.780)	(-4.17)	(-3.055)	(-3.57)
DE	0.000	0.0416***	−0.001	0.0566***
	(1.053)	(5.22)	(-1.364)	(8.88)
ROA	0.283	0.1335	0.404	0.3158
	(0.961)	(0.79)	(1.479)	(1.15)
RET	0.170***	0.1584**	0.042	0.1067**
	(5.725)	(3.19)	(1.350)	(2.73)
Std5RET	−0.724	−2.2751***	−2.160***	−2.4650***
	(-1.264)	(-3.54)	(-4.151)	(-4.14)
Std5ROA	0.000	−0.5749*	0.016**	−0.606
	(1.029)	(-1.96)	(2.182)	(-1.15)
Constant	5.729***	7.1183***	7.604***	7.6869***
	(27.978)	(63.10)	(26.281)	(59.86)
Chow test of	(1)–(3)	(2)–(4)		
[Prob > F]	[0.8188]	[0.0344]**		
Control *for*				
Industry FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Obs.	9,673	7,879	7,201	2,171
Adj. R^2	0.265	0.397	0.127	0.485

personal use of social media can act as a strategic tool when competing with other firms. Our results in Tables 5 and 6 also provide a new driving force that links CEO compensation to CEO characteristics. In addition to CEO's managerial ability that reduces agency costs, we indicate that a firm awards CEO's efforts in social media because these efforts help when this firm faces higher information asymmetry or greater market competition (Demerjian, Lev, & McVay, 2012).

4.4. CEO Twitter usage and earnings announcement responses

The abnormal returns around earnings announcements well reflect market efficiency that new information changes investors' expectations in stocks (May, 1971; Kane et al., 1984; La Porta et al., 1997; Fama, 1998; Kaniel et al., 2012). However, Lee et al. (2015) showed that the existence of Twitter use by a firm could weaken the negative price reaction to recall announcements by directly broadcasting the firm's intended message to investors without distorting the content. When this information is properly managed, firms can communicate with users and customers through social media to mitigate the adverse effects related to unpleasant corporate

Table 6
CEO Twitter usage and product market competition
This table presents the results of the OLS estimation of the influence of CEO Twitter usage on their compensation by considering the subsample of market competition. We use several proxies for market competition such as consumer product-oriented (CP oriented), product similarity, and product market fluidity (Hoberg and Phillips (2016), Hoberg et al. (2014), and Hoberg and Phillips (2010)).
The equation is as follows:

$$TDC_{i,t} = \alpha_0 + \alpha_1 CEO_Twitter_{i,t} + \beta' F_{i,t-1} + \theta' Z_{i,t} + \gamma_i + \mu_t + \varepsilon_{i,t}$$

where the dependent variable is *TDC* (total compensation) and the main explanatory variable is *CEO_Twitter*. Specifically, $TDC_{i,t}$ is the natural logarithm of compensation for CEO *i* in year *t*; *CEO_Twitter* is an indicator for CEOs who use Twitter that equals one when CEOs have Twitter accounts, and zero otherwise. $F_{i,t-1}$ is a vector of control variables (firm characteristics) for firm i in year *t*-1. $Z_{i,t}$ is the vector of the control variables for corporate governance and CEO characteristics *i* in year *t*. The γ_i and μ_t represent the fixed effects of industry and year, respectively, and $\varepsilon_{i,t}$ represents the error term of the regression. In all specifications, the *t-statistics* reported are based on heteroscedasticity among the firm-level, robust standard errors (White, 1980). Appendix A provides the definitions of variables. The *, **, and *** denote the significance levels of 10%, 5%, and 1%, respectively.

	High product market competition			Low product market competition		
	Similarity ≥ Median	Fluidity ≥ Median	CP oriented < Median	Similarity < Median	Fluidity < Median	CP oriented ≥ Median
	(1)	(2)	(3)	(4)	(5)	(6)
CEO_Twitter	0.252***	0.184**	0.402***	0.004	0.105	0.059
	(2.753)	(1.974)	(3.048)	(0.047)	(1.397)	(0.773)
CEO5pct	−0.229***	−0.293***	−0.287***	−0.266***	−0.208***	−0.231***
	(-3.362)	(-4.420)	(-3.954)	(-5.542)	(-4.833)	(-5.205)
CEO_firstyear	−0.029	−0.010	0.014	−0.042	−0.071*	−0.059
	(-0.609)	(-0.199)	(0.212)	(-0.936)	(-1.703)	(-1.574)
CEO_tenure	0.026	0.071*	0.047	0.069*	0.024	0.045
	(0.690)	(1.855)	(0.922)	(1.763)	(0.699)	(1.393)
CEO_chair	0.219***	0.230***	0.120	0.168***	0.126***	0.194***
	(4.328)	(4.675)	(1.639)	(3.569)	(3.106)	(5.054)
Bdsize	0.125***	0.124***	0.152***	0.148***	0.150***	0.132***
	(8.431)	(7.301)	(9.792)	(7.764)	(12.217)	(8.583)
Indcompcom	0.300***	0.389***	0.476***	0.285***	0.194**	0.259***
	(4.109)	(4.502)	(3.742)	(3.139)	(2.577)	(3.842)
BMV	−0.379***	−0.313***	−0.134	−0.199	−0.205*	−0.351***
	(-3.624)	(-2.830)	(-0.681)	(-1.631)	(-1.722)	(-4.248)
DE	−0.002**	−0.000	0.000	0.000	0.002***	−0.000
	(-2.071)	(-0.970)	(0.502)	(1.188)	(2.919)	(-0.172)
ROA	0.884***	0.382*	0.538	0.067	0.845**	0.413*
	(3.839)	(1.678)	(1.035)	(0.192)	(2.173)	(1.876)
RET	0.160***	0.137***	0.083*	0.132***	0.168***	0.164***
	(6.099)	(5.482)	(1.796)	(5.058)	(6.322)	(7.917)
Std5RET	−1.316***	−1.046***	−1.291*	−2.593***	−3.452***	−1.505***
	(-4.394)	(-3.547)	(-1.693)	(-3.222)	(-4.384)	(-4.063)
Std5ROA	0.001***	0.001***	0.000	0.000	0.000	0.001**
	(2.907)	(2.906)	(0.967)	(0.136)	(0.518)	(2.382)
Constant	6.939***	6.881***	6.993***	7.221***	7.457***	7.476***
	(15.779)	(21.782)	(24.067)	(29.354)	(37.998)	(29.478)
Chow test of	(1)–(4)	(2)–(5)	(3)–(6)			
[Prob > F]	[0.0446]**	[0.5379]	[0.0261]**			
Control for						
Industry FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Obs.	8,309	8,257	3,742	8,298	8,252	12,942
Adj. R^2	0.196	0.187	0.214	0.229	0.225	0.194

events.

To examine the effect of CEO Twitter usage on earnings announcements, we focus on the cumulative abnormal return (*CAR*) and run a regression model as follows:

$$CAR[-1;1]_{i,t} = \beta_1 SUE_{i,t} + \beta_2 CEO_Twitter_{i,t} + \beta_3 SUE_{it} \times CEO_Twitter_{i,t} + \theta' Z_{i,t-1} + u_j + v_t + \varepsilon_{i,t}, \quad (2)$$

where $CAR[-1;1]_{i,t}$ is the cumulative abnormal return (from the Fama-French three-factor model)[7] in the window [-1;1]; $SUE_{i,t}$ is the difference between the actual value of earnings per share (EPS) and the mean (median) of values forecasted by analysts that is divided by the standard deviation in the EPS forecast; $Z_{i,t-1}$ is the vector of control variables; u_j and v_t are industry and year fixed effects,

[7] The main results remained if we adjusted the returns with the Carhart 4-factor model.

respectively. The control variables are *Assets* (natural logarithm of total assets), *Q* (Tobin's Q), *Leverage* (long-term debt plus debt in current liabilities all divided by total assets), *BHAR* (buy-and-hold abnormal return), *Tangibility* (net property, plant, and equipment divided by total assets), *Profitability* (earnings before interest, taxes, depreciation, and amortization divided by total assets), *Z-score* (modified Altman's Z-score), and *CF volatility* (standard deviation in quarterly operating cash flows over the four fiscal years).

Our key independent variable is the interaction between *SUE_mean*× *CEO_Twitter* and *SUE_median*× *CEO_Twitter* in which *SUE* reflects the information asymmetry between analysts. We predict a negative coefficient for the interaction term when the CEO's Twitter usage lowers the information asymmetry around the earnings announcements.

Table 7 presents the results. First, *SUE* positively and significantly affects the abnormal returns around earnings announcements whether we use the mean or median of analysts' forecasts to estimate it. Second, the coefficients for the interaction terms (i.e., *SUE_mean*× *CEO_Twitter* and *SUE_median*× *CEO_Twitter*) are negative in columns (1) and (2). Although the CEO's use of Twitter cannot fully alleviate the effects of information asymmetry on stock returns, the findings suggest that CEO Twitter usage helps reduce information uncertainty and alleviates future return drift around the earnings announcements.

The results in Table 7 support H4 that *the effect of CEO Twitter usage is more likely to curtail the market responses to earnings announcements*. These findings further echo our results in Table 5 by showing that the CEO's Twitter usage reduces information asymmetry. When analysts have a larger deviation in a firm's EPS forecast, the CEO's Twitter helps reduce the price reaction due to information asymmetry.

5. Additional supporting evidence

5.1. Endogeneity

In this subsection, we address endogeneity, which is the central concern surrounding most empirical studies in the finance field. Because Twitter plays an important role in directly disseminating and updating information to the public, its use is undoubtedly a managerial decision. Malhotra and Malhotra (2016) argued that CEOs might initiate and influence online conversations via tweeting. They pointed out that CEOs could use Twitter to increase influence, create positive sentiment, or spread a message to the public. Therefore, we need to address other omitted variables that might drive the reasons why a CEO uses Twitter.

To address the omitted variable concerns, we use an instrumental variable to represent a rival's Twitter usage to identify a CEO's incentive for using Twitter. Appendix C gives an example of a CEO who was followed by their rival's CEO on Twitter. In the airplane industry, Denis Muilenburg (CEO of Boeing) and Guillaume Faury (CEO of Airbus) follow each other, and both follow Tory Bruno (CEO of United Launch Alliance). We find similar results in the technology industry; Tim Cook (CEO of Apple) follows Sunder Pichai (CEO of Google) and Satya Nadella (CEO of Microsoft); Marissa Mayer (CEO of Yahoo) follows Jeremy Stoppleman (CEO of Yelp) and Sundar Pichai; Sundar Pichai follows Marissa Mayer, Jeff Bezos (CEO of Amazon), Satya Nadella, and Tim Cook. In addition to the reasons shown in Malhotra and Malhotra (2016), we argue that CEOs may use Twitter because their competitors use it. By using Twitter, a CEO is able to update their competitors' information and express their affinity for them (Malhotra & Malhotra, 2016).

Following this natural design in Twitter, we use the rival CEO's Twitter usage (*Rival_Twitter*) as a proxy for a CEO's Twitter usage. *Rival_Twitter* is the percentage of rival CEOs who use Twitter in the corresponding year. We identify rival CEOs in the same industry by the proportion of them who use Twitter based on the Fama and French 12-industry classification (ffsic12) or Fama and French 48-industry classification (ffsic 48).

By using a two-stage least-square (2LSL) regression, we first regress *CEO_Twitter* on *Rival_Twitter* and other exogenous variables as follows:

$$CEO_Twitter_{i,t} = \delta Rival_Twitter_{i,t} + \theta' Z_{i,t} + u_j + v_t + \varepsilon_{i,t}, \tag{3}$$

where *Rival_Twitter* for firm i at time t is the percentage of rivals' CEOs in the same industry who use Twitter in year t. $Z_{i,t}$ refers to control variables for firm i at time t such as assets, age, Tobin's Q, leverage, tangibility, and profitability. The u_j and v_t capture industry and year effects, respectively.

In the second stage, we use the fitted value obtained from the first stage in the regression model as follows:

$$TDC_{i,t} = \beta \widehat{CEO_Twitter}_{i,t-1} + \theta' Z_{i,t-1} + u_j + v_t + \varepsilon_{i,t}, \tag{4}$$

where $TDC_{i,t}$ is the natural logarithm of compensation for a CEO; $\widehat{CEO_Twitter}_{i,t-1}$ is the fitted value obtained from the first stage for firm i at time t-1. $Z_{i,t-1}$ refers to the control variables for firm i at time t-1.

Table 8 shows the results. The results in columns (1) and (3) indicate that the percentage of the use of Twitter by a rival's CEO well explains a CEO's tendency to use Twitter. This finding supports our prediction that CEOs are likely to follow competitors when deciding whether to use social media such as Twitter. Twitter usage in the corresponding industry can well predict a CEO's incentive in using Twitter.

In terms of CEO compensation, the results in columns (2) and (4) show that the instrumental variables significantly and positively affect a CEO's total compensation. By focusing on Twitter coverage in the 48-industry classification, we find a stronger effect of CEO Twitter usage on compensation. These results help address the concerns related to the omitted reasons for using Twitter.

Table 7
CEO Twitter usage and earnings announcement responses
This table presents the regression results for the effects of using Twitter on the cumulative abnormal return (*CAR*) when we use the sensitivity to earnings announcements. We run a regression model as follows:

$$CAR[-1;1]_{it} = \beta_1 SUE_{it} + \beta_2 CEO_Twitter_{it} + \beta_3 SUE_{it} \times CEO_Twitter_{it} + \boldsymbol{\gamma}' \boldsymbol{Z}_{it-1} + u_j + \nu_t + \varepsilon_{it}$$

where $CAR[-1;1]_{it}$ is the cumulative abnormal return (from the Fama-French 3-factor model) in the window [-1;1]; SUE_{it} is the difference between the actual EPS and the mean (median) forecasted EPS that is scaled by the standard deviation of the forecasted EPS by analysts; $SUE_{it} \times CEO_Twitter_{it}$ is the interaction between *SUE* and *CEO_Twitter;* $\boldsymbol{Z}_{it-1}$ is the vector of control variables; u_j and ν_t are industry and year fixed effects, respectively. The control variables are *Asset* (natural logarithm of total assets), *Q* (Tobin's Q), *Leverage* (ratio of long-term debt plus debt in current liabilities to total assets), *BHAR* (the corresponding buy-and-hold abnormal return), *Tangibility* (ratio of net property, plant, and equipment to total assets), *Profitability* (ratio of earnings before interest, taxes, depreciation, and amortization to total assets), *Z-score* (a modified Altman's Z-score), and *CF volatility* (ratio of standard deviation of quarterly cash flows from operations over the four fiscal years prior to the loan initiation year to total debt). Appendix A provides the definitions of variables. In all equations, we report *t*-values based on the robust standard errors adjusted for clustering at the firm level (White,1980). Appendix A provides the definitions of variables. Superscripts *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)
CEO_Twitter	0.0016	0.0016
	(0.49)	(0.48)
SUE_mean	0.0026***	
	(8.08)	
SUE_mean × *CEO_Twitter*	−0.0011**	
	(-2.12)	
SUE_median		0.0025***
		(8.09)
SUE_median × *CEO_Twitter*		−0.0011*
		(-1.94)
Assets	−0.0017**	−0.0018**
	(-3.05)	(-3.05)
Q	−0.0022	−0.0022
	(-1.70)	(-1.70)
Leverage	0.0088	0.0089
	(1.18)	(1.21)
BHAR	−0.0047	−0.0047
	(-1.13)	(-1.12)
Tangibility	−0.0172	−0.0172
	(-1.22)	(-1.22)
Profitability	−0.0148	−0.0149
	(-0.44)	(-0.44)
Z-score	−0.0001	−0.0001
	(-0.09)	(-0.08)
CF volatility	0.000	0.000
	(-0.23)	(-0.23)
Constant	0.0284***	0.0285***
	(3.57)	(3.57)
Control for		
Industry FE	YES	YES
Year FE	YES	YES
Obs.	6,065	6,065
Adj R^2	0.037	0.037

5.2. Placebo test

In the baseline regression model, we discuss CEO Twitter usage by focusing on the relationship between compensation and the Twitter indicator in the same year. This setting involves one potential concern; there could be other confounding factors that happen simultaneously. To alleviate this concern, we use alternative models that replace our key variable of interest with measures that are 3-year, 2-year, or 1-year-lagged. We expect non-positive coefficients for these alternative measures because CEOs do not use Twitter in these alternative windows. In this case, CEOs should not have higher compensation when there is no Twitter usage to help reduce information asymmetry.

Table 9 shows the results. We find consistent and negative relationships between our alternative measures and total compensation. In terms of statistical significance, the results show that all alternative measures have coefficients that are indifferent to zero. This alternative setting shows that CEOs who use Twitter receive higher compensation only after they establish that usage. Therefore, the concern of confounding factors does not affect our results.

Table 8
Two-stage least squares (2SLS) regression.
This table provides the results of the two-stage least-squares (2SLS) regression for the effect of CEO Twitter usage on their compensation (*TDC*).
Stage 1: Regress *CEO_Twitter* on instrument variable *Rival_Twitter* and other exogenous variables as Equation (1):

$$CEO_Twitter_{i,t} = \delta Rival_Twitter_{i,t} + \theta' Z_{i,t} + u_j + v_t + \varepsilon_{i,t}$$

Stage 2: Replace the fitted value for *CEO_Twitter* from the first stage of the model in the main regression with this equation:

$$TDC_{i,t} = \beta \widehat{CEO_Twitter}_{i,t} + \theta' Z_{i,t-1} + u_j + v_t + \varepsilon_{i,t}$$

where $TDC_{i,t}$ is the natural logarithm of compensation for CEO *i* in year *t*; $CEO_Twitter_{i,t}$ is an indicator for CEOs who use Twitter that equals one when CEOs have Twitter accounts, and zero otherwise. $\widehat{CEO_Twitter}_{i,t-1}$ is the fitted value from the first-stage regression (2) for firm *i* at time *t*-1. $CEO_Twitter_{i,t}$ represents a CEO with a Twitter account equals one if the CEO of firm *i* adopted Twitter after the year *t*, and zero otherwise. $Rival_Twitter_{i,t}$ for firm *i* at time *t* is the percentage of rivals in the same industry that follow the CEO's Twitter in year *t*. $Z_{i,t-1}$ refers to control variables for firm *i* at time *t*-1. The u_j and v_t capture industry effect and year effects, respectively. Appendix A provides the definitions of variables. Superscripts *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Model 1: 12-industry classifications		Model 2: 48-industry classifications	
	CEO_Twitter	TDC	CEO_Twitter	TDC
	(1)	(2)	(3)	(4)
Rival_Twitter	0.647***		0.162***	
	(4.408)		(2.613)	
CEO_twitter		0.268*		0.696*
		(1.749)		(1.747)
CEO5pct	−0.031	−0.235***	−0.049*	−0.191***
	(-1.373)	(-4.269)	(-1.942)	(-3.152)
CEO_firstyear	0.000	−0.043	−0.006	−0.058
	(0.010)	(-0.991)	(-0.449)	(-1.361)
CEO_tenure	0.006	−0.007	0.009	0.015
	(0.390)	(-0.184)	(0.567)	(0.423)
CEO_chair	−0.022	0.147***	−0.027	0.211***
	(-1.243)	(3.107)	(-1.485)	(4.434)
Bdsize	0.012***	0.118***	0.010**	0.121***
	(2.916)	(8.147)	(2.191)	(7.522)
Indcompcom	−0.035	0.390***	−0.028	0.349***
	(-0.989)	(4.990)	(-0.759)	(4.188)
BMV	−0.032	−0.498***	−0.141***	−0.128
	(-0.638)	(-4.598)	(-2.607)	(-0.981)
DE	0.000**	0.000	0.000***	0.000
	(1.968)	(0.933)	(2.602)	(0.090)
ROA	−0.075	0.677*	0.244	−4.371***
	(-0.534)	(1.878)	(0.771)	(-4.026)
RET	0.005	0.169***	0.008	0.215***
	(0.455)	(6.016)	(0.772)	(7.393)
Std5RET	0.000	0.000	0.000	0.000
	(0.736)	(1.625)	(0.262)	(1.483)
Std5ROA	−0.223	−3.829***	0.244	−4.371***
	(-0.810)	(-3.915)	(0.771)	(-4.026)
Control for				
Industry FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Obs.	11,240	11,237	11,244	11,241
Adj R^2	0.266	0.105	0.107	0.157

5.3. Alternative measures of CEO compensation

We also examine the baseline regression model by using alternative measures for CEO compensation. Specifically, we replace *TDC* with the following variables: *Delta*, *Vega*, *Firm-related wealth*, and *non-equity-based*. *Delta* stands for a CEO's change in wealth (expressed in thousands of dollars) for a 1% increase in the firm's stock price. We use the information on CEOs' complete portfolios of stock and options from Execucomp or identify it by using the methodology from Yermack (1995) and Core and Guay (2002). The value of the option portfolio is the sum of the Black–Scholes values of the newly granted options and previously granted unexercisable and exercisable options. The Yermack's (1995) sample contains information on a CEO's previously granted stock but not options, so we used information on the options granted only in the last year to calculate the delta of the CEOs' options. *Vega* stands for the CEO's change in wealth (expressed in thousands of dollars) for a 1% increase in the annualized standard deviation in the firm's stock returns. *Firm-related wealth* is the sum of the value of the stock and option portfolio held by the CEO (Daniel, Li, & Naveen, 2013), and *non-equity-based* compensation is the total compensation (*TDC*) minus equity-based compensation (Chhaochharia & Grinstein, 2009).

Compared with CEO raw compensation, *Delta* and *Vega* help identify how a CEO's compensation is related to a firm's stock performance. *Delta* and *Vega* also imply whether a CEO has an incentive to take risks. *Firm-related wealth* focuses on equity-based compensation, while *non-equity-based* identifies only cash-based compensation.

Table 9
Placebo test
This table presents the results from regressing *TDC* (total compensation) on the 3-year-lagged *CEO_Twitter* (in Model 1); 2-year-lagged *CEO_Twitter* (in Model 2), and the 1-year-lagged *CEO_Twitter* (in Model 3). $Z_{i,t}$ is the vector of the control variables for corporate governance and CEO characteristics i in year t. The γ_i and μ_t represent the fixed effects of industry and year, respectively, and $\varepsilon_{i,t}$ represents the error term of the regression. In all specifications, the *t-statistics* reported are based on heteroscedasticity among the firm-level, robust standard errors (White, 1980). Appendix A provides the definitions of variables. The *, **, and *** denote the significance levels of 10%, 5%, and 1%, respectively.

	(1)	(2)	(3)
	TDC	TDC	TDC
$CEO_Twitter_{t-3}$	−0.098		
	(-1.159)		
$CEO_Twitter_{t-2}$		−0.095	
		(-1.052)	
$CEO_Twitter_{t-1}$			−0.067
			(-0.673)
CEO5pct	−0.255***	−0.255***	−0.256***
	(-6.288)	(-6.295)	(-6.313)
CEO_firstyear	−0.045	−0.045	−0.046
	(-1.340)	(-1.345)	(-1.358)
CEO_tenure	0.044	0.044	0.044
	(1.527)	(1.527)	(1.537)
CEO_chair	0.179***	0.179***	0.179***
	(4.979)	(4.975)	(4.969)
Bdsize	0.139***	0.139***	0.139***
	(10.580)	(10.582)	(10.582)
Indcompcom	0.300***	0.300***	0.299***
	(4.877)	(4.871)	(4.851)
BMV	−0.296***	−0.297***	−0.299***
	(-3.458)	(-3.471)	(-3.503)
DE	−0.000	−0.000	−0.000
	(-0.242)	(-0.242)	(-0.224)
ROA	0.420**	0.419**	0.413*
	(1.976)	(1.970)	(1.947)
RET	0.150***	0.149***	0.150***
	(8.001)	(8.003)	(8.007)
Std5ROA	−1.506***	−1.507***	−1.509***
	(-4.296)	(-4.299)	(-4.305)
std5RET	0.000	0.000	0.000
	(0.966)	(0.966)	(0.972)
Constant	7.333***	7.329***	7.299***
	(33.362)	(33.141)	(32.320)
Control for			
Industry FE	YES	YES	YES
Year FE	YES	YES	YES
Obs.	16,684	16,684	16,684
Adj R^2	0.192	0.192	0.192

Table 10 presents the results. Across all specifications, the coefficients for *CEO_Twitter* are significantly and positively related to *Delta*, *Vega*, *Firm-related wealth*, and *non-equity-based*. Thus, these results provide further support for H1. According to the results in columns (3) and (4), CEO Twitter usage has a stronger effect on equity-based compensation measured by *Firm-related wealth*.

5.4. Firm fixed effects

In the baseline results, we control for industry and year-fixed effects to ensure that our results are not attributed to the heterogeneity between industries and years. Although $CEO_Twitter_{i,t}$ can capture the Twitter users' own difference in compensation before and after the use of Twitter, the results still could reflect the heterogeneity between CEOs. We further control for firm and year-fixed effects to capture omitted variables driving the association between Twitter users and compensation.

Table 11 presents the results. We find that *CEO_Twitter* positively and significantly affects *TDC* after controlling for all potential factors, firm-fixed effects, and year-fixed effects. Compared with *CEO_Twitter*, other CEO characteristics such as *CEO5pct* and *CEO_chair* have weaker effects on compensation after controlling for firm-fixed effects. The effects of board structure on CEO compensation also become weaker when firm-fixed effects are added. Overall, the positive relationship between the CEO's Twitter usage and compensation remains after controlling for firm-fixed effects.

Table 10
Alternative measures of CEO compensation
This table presents the regression results for the effect of CEO Twitter usage on alternative measures of compensation.
The equation is as follows:

$$Compensation_{i,t} = \alpha_0 + \alpha_1 CEO_Twitter_{i,t} + \beta' F_{i,t-1} + \theta' Z_{i,t} + \gamma_i + \mu_t + \varepsilon_{i,t}$$

where the dependent variable is *Compensation* (*Delta, Vega, Firm related wealth, Non-equity-based*) and the main explanatory variable is *CEO_Twitter*. Specifically, *Compensation* is the natural logarithm of delta, vega, firm related wealth, non-equity-based alternatively for CEO *i* in year *t*; $CEO_Twitter_{i,t}$ is an indicator for CEOs who use Twitter that equals one when CEOs have Twitter accounts, and zero otherwise. $F_{i,t-1}$ is a vector of control variables (firm characteristics) for firm *i* in year *t*-1. $Z_{i,t}$ is the vector of the control variables for corporate governance and CEO characteristics *i* in year *t*. The γ_i and μ_t represent the fixed effects of industry and year, respectively, and $\varepsilon_{i,t}$ represents the error term of the regression. In all specifications, the *t-statistics* reported are based on heteroscedasticity among the firm-level, robust standard errors (White, 1980). Appendix A provides the definitions of variables. The *, **, and *** denote the significance levels of 10%, 5%, and 1%, respectively.

	(1)	(2)	(3)	(4)
	Delta	*Vega*	*Firm-related wealth*	*Non-equity-based*
CEO_Twitter	0.339***	0.295**	0.313***	0.218***
	(4.116)	(2.312)	(3.772)	(4.159)
CEO5pct	0.654***	−0.097	0.762***	−0.270***
	(10.623)	(-0.740)	(12.512)	(-7.978)
CEO_firstyear	−0.633***	−0.532***	−0.605***	−0.060*
	(-12.850)	(-5.652)	(-12.502)	(-1.776)
CEO_tenure	0.623***	0.132	0.653***	0.055*
	(12.451)	(1.531)	(12.966)	(1.892)
CEO_chair	0.652***	0.440***	0.662***	0.205***
	(12.423)	(4.582)	(12.619)	(6.252)
Bdsize	0.134***	0.245***	0.120***	0.139***
	(8.984)	(8.114)	(8.292)	(10.434)
Indcompcom	0.343***	0.555*	0.315***	0.339***
	(3.435)	(1.940)	(3.057)	(4.579)
BMV	−2.010***	−1.045***	−1.967***	−0.491***
	(-17.807)	(-4.559)	(-17.333)	(-6.472)
DE	−0.000**	−0.001	−0.001**	−0.000
	(-2.361)	(-1.247)	(-2.480)	(-0.290)
ROA	0.054	0.500	0.034	0.028
	(0.182)	(0.881)	(0.114)	(0.137)
RET	0.675***	0.277***	0.702***	0.170***
	(9.117)	(4.781)	(9.274)	(8.329)
Std5RET	−2.474***	−2.397**	−1.967***	−1.591***
	(-4.739)	(-2.098)	(-3.864)	(-4.693)
Std5ROA	−0.001	−0.003***	−0.001	0.000
	(-1.475)	(-6.026)	(-1.142)	(0.939)
Constant	3.670***	2.995***	7.864***	7.245***
	(14.715)	(5.632)	(31.766)	(38.928)
Control for				
Industry FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Obs.	12,342	9,811	12,342	12,398
Adj. R^2	0.393	0.105	0.397	0.287

5.5. Twitter usage and revenue efficiency

In this subsection, we examine whether CEO's Twitter usage is related to CEO performance. In addition to the role of Twitter usage in reducing information asymmetry and market competition, we would like to know if Twitter usage also helps improve firm performance. The relationship between Twitter usage and firm performance also helps clarify a broader association between Twitter usage and compensation. A Twitter user may have a better ability not only to reduce external information asymmetry but also to enhance internal productivity.

In order to identify the CEO's own effects on firm performance, we follow Demerjian et al. (2012) to estimate the revenue-generating efficiency and managerial ability of firms as follows:

$$Revenue\ efficiency_i = \alpha + \beta_1 Size_i + \beta_2 Market\ share_i + \beta_3 Free\ cash\ flow_i + \beta_4 Ln(Age)_i + \beta_5 Bussiness\ Segment\ Concentration_i + \beta_6 Foreign\ Currency\ indicator_i + Year_i + \varepsilon_i, \quad (5)$$

where *Revenue efficiency* is estimated by data envelopment analysis, which captures how different investments generate revenue in specific industries. The residual of Equation (5) measures *Managerial ability*. A firm is more productive when its revenue efficiency is high, while managerial ability is the productivity that is less likely to come from firm characteristics or other confounding factors.

Table 12 presents the results. We indicate a positive effect of *CEO_Twitter* on *Revenue efficiency* in columns (1) and (2). The results indicate a broader relationship between CEO Twitter usage and compensation by showing that CEOs with better external image also

Table 11
Alternative empirical design – Firm-fixed effect
These tables present the results of the ordinary least squares (OLS) of CEO Twitter usage on their compensation.
$TDC_{i,t} = \alpha_0 + \alpha_1 CEO_Twitter_{i,t} + \beta' F_{i,t-1} + \theta' Z_{i,t} + \vartheta_i + \mu_t + \varepsilon_{i,t}$
where the dependent variable is TDC (total compensation) and the main explanatory variable is *CEO_Twitter*. Specifically, $TDC_{i,t}$ is the natural logarithm of compensation for CEO *i* in year *t*; *CEO_Twitter* is an indicator for CEOs who use Twitter that equals one when CEOs have Twitter accounts, and zero otherwise. $F_{i,t-1}$ is a vector of control variables (firm characteristics) for firm *i* in year *t*-1. $Z_{i,t}$ is the vector of the control variables for corporate governance and CEO characteristics *i* in year *t*. ϑ_i and μ_t represent the fixed effects of firm and year, respectively, and $\varepsilon_{i,t}$ represents the error term of the regression. Other variables are defined in Appendix A. In all specifications, the *t-statistics* reported are based on heteroscedasticity among the firm-level, robust standard errors (White, 1980). The *, **, and *** denote the significance levels of 10%, 5%, and 1%, respectively.

	(1)	(2)	(3)
	TDC	TDC	TDC
CEO_Twitter	1.550***	1.495***	1.229***
	(60.289)	(36.113)	(22.124)
CEO5pct		−0.011	−0.013
		(-0.289)	(-0.329)
CEO_firstyear		0.014	0.016
		(0.427)	(0.475)
CEO_tenure		0.136***	0.122***
		(3.669)	(3.299)
CEO_chair		0.020	0.014
		(0.451)	(0.331)
Bdsize		0.011	0.014*
		(1.515)	(1.951)
Indcompcom		−0.002	0.011
		(-0.048)	(0.324)
BMV			−0.470***
			(-7.382)
DE			−0.000
			(-0.647)
ROA			0.197
			(1.143)
RET			0.158***
			(10.409)
Std5ROA			−0.389
			(-1.180)
Std5RET			0.000**
			(2.476)
Constant	6.554***	6.154***	6.821***
	(47.309)	(32.938)	(29.865)
Control for			
Firm FE	YES	YES	YES
Year FE	YES	YES	YES
Obs.	16,684	16,684	16,684
Adj R^2	0.665	0.666	0.671

help improve a firm's internal productivity. After identifying the CEO's own ability, we also find that Twitter users are likely to have a higher *Managerial ability*. Therefore, CEOs who use Twitter may benefit from internal productivity by improving revenue efficiency. Our results are consistent with Kirtiş and Karahan (2011), who show that social media helps improve the communication between firms and stakeholders.

6. Conclusion

Our study is the first to show how CEOs' use of Twitter can affect their compensation. With the complete list of all CEOs in Execucomp, we identify users with active Twitter accounts by those with the same first and last names as the CEO in firm-related information. Through this labor-intensive process, we identify 336 CEOs that have active personal Twitter accounts and firm characteristics so that we could conduct our tests.

Using an S&P 1500 sample from 2006 to 2020, we find that CEOs receive higher compensation after they create a Twitter account, confirming that involvement in social media helps CEOs increase their bargaining power. We further clarify the underlying channels of this CEO's Twitter usage through information asymmetry and market competition. We also examine the consequences of CEO Twitter usage by showing that this usage helps curtail the stock price deviation around earning announcements and improves revenue efficiency. Alternative settings for the instrumental variables, a placebo test, alternative measures of CEO compensation, and alternative empirical designs verified our results.

Overall, our findings support the idea that CEOs who use Twitter receive higher compensation because social media helps reduce

Table 12
CEO Twitter usage and revenue efficiency
These tables present the results of the ordinary least squares (OLS) of CEO Twitter usage on the firm efficiency.
$Efficiency_{i,t} = \alpha_0 + \alpha_1 CEO_Twitter_{i,t} + \beta' F_{i,t-1} + \theta' Z_{i,t} + \gamma_i + \mu_t + \varepsilon_{i,t}$
where the dependent variable is *Efficiency* and the main explanatory variable is *CEO_Twitter*. Specifically, $Efficiency_{i,t}$ is firm efficiency such as revenue efficiency and managerial ability in year *t*; *CEO_Twitter* is an indicator for CEOs who use Twitter that equals one when CEOs have Twitter accounts, and zero otherwise. $F_{i,t-1}$ is a vector of control variables (firm characteristics) for firm *i* in year *t*-1. $Z_{i,t}$ is the vector of the control variables for corporate governance and CEO characteristics *i* in year *t*. γ_i and μ_t represent the fixed effects of industry and year, respectively, and $\varepsilon_{i,t}$ represents the error term of the regression. Other variables are defined in Appendix A. In all specifications, the *t-statistics* reported are based on heteroscedasticity among the firm-level, robust standard errors (White, 1980). The *, **, and *** denote the significance levels of 10%, 5%, and 1%, respectively.

Dependent variable	*Revenue efficiency*		*Managerial ability*	
	(1)	(2)	(3)	(4)
CEO_Twitter	0.077***	0.064***	0.048***	0.041***
	(5.264)	(5.172)	(4.399)	(4.097)
CEO5pct		−0.061***		−0.027***
		(-6.913)		(-4.064)
CEO_firstyear		−0.001		0.004
		(-0.110)		(0.570)
CEO_tenure		0.002		0.003
		(0.327)		(0.698)
CEO_chair		0.013*		0.000
		(1.866)		(0.069)
Bdsize		0.024***		0.010***
		(12.908)		(6.296)
Indcompcom		0.026*		0.008
		(1.804)		(0.660)
BMV		−0.126***		−0.112***
		(-7.311)		(-7.524)
DE		−0.000		−0.000*
		(-0.758)		(-1.839)
ROA		0.229***		0.191***
		(4.631)		(4.468)
RET		0.024***		0.021***
		(4.873)		(5.477)
Std5ROA		0.204**		0.330***
		(2.450)		(4.399)
Std5RET		0.000		0.000
		(0.236)		(0.211)
Constant	0.375***	0.086***	−0.233***	−0.354***
	(30.186)	(2.739)	(-19.597)	(-13.376)
Control for				
Industry FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Obs.	10,424	10,424	10,424	10,424
Adj R^2	0.193	0.324	0.146	0.229

information asymmetry between firms and investors. Our results shed light on the growing influence of social media in financial markets and show that its usage can raise CEOs' bargaining power in salary negotiations.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Appendix A. Definitions of variables

Variables	Definition	Data source
CEO compensation variable		

(*continued on next page*)

(*continued*)

Variables	Definition	Data source
Total annual compensation (*TDC*)	Total compensation consists of salary, bonus, value of restricted stock granted, value of options granted (using Black-Scholes), long-term incentive payouts, and other compensation.	Execucomp
Twitter variables		
CEO_Twitter	An indicator for CEOs who use Twitter that equals one when CEOs have Twitter accounts, and zero otherwise.	Hand collect
Rival_Twitter	Percentage of other CEOs in the same industry that uses Twitter in year *t*.	Hand collect
Firm characteristics		
Assets	Natural logarithm of total assets.	Compustat
BHAR	Buy and hold abnormal return, $BHAR_{iT} = \prod_{t=1}^{T}(1+R_{it}) - \prod_{t=1}^{T}(1+R_{mt})$, where R_{it} is the monthly stock return of firm i and R_{mt} is the monthly value-weighted CRSP market return. *BHAR3y* and *BHAR5y* are estimated for 3-year and 5-year periods, respectively.	CRSP
BMV	Book-to-market ratio.	Compustat
CF volatility	Standard deviation of quarterly operation cash-flows over the four fiscal years scaled by total assets.	Compustat
DE	The total liabilities divided by total equity.	Compustat
Leverage	Long-term debt plus debt in current liabilities divided by total assets.	Compustat
Managerial ability	Residual of equation model that regresses efficiency on the size, market share, free cash flow, age, business segment concentration, foreign currency indicator, and year dummies of firms (Demerjian et al., 2012).	Demerjian et al. (2012)
Profitability	Earnings before interest, taxes, depreciation, and amortization (EBITDA) divided by total assets. This measure is adjusted by the industry median of profitability.	Compustat
Q	Tobin's Q.	Compustat and CRSP
RET	Firm's annual common stock return.	Compustat
Revenue efficiency	Revenue efficiency is estimated by the data envelopment analysis that captures how different investments generate revenue in a specific industry (Demerjian et al., 2012).	Demerjian et al. (2012)
ROA	Return on assets.	Compustat
Std5RET	Standard deviation in common stock returns over the previous five years.	Compustat
Std5ROA	Standard deviation in return on assets over the previous five years.	Compustat
Tangibility	Net property, plant, and equipment divided by total assets.	Compustat
Z-score	Modified Altman's Z-score (1.2 × working capital + 1.4 × retained earnings + 3.3 × EBIT + 0.999 × sales)/Total assets.	Compustat
CEO characteristics and corporate governance		
Bdsize	Number of board directors.	Execucomp
CEO5pct	An indicator variable that equals one if the CEO owns 5% or more of the firm's stock, and zero otherwise.	Execucomp
CEO_firstyear	An indicator variable that equals one if it is the CEO's first year of service at a firm, and zero otherwise.	Execucomp
CEO_tenure	The natural logarithm of CEO tenure.	Execucomp
CEO_chair	An indicator variable that equals one when the CEO is also the board chairperson and zero otherwise.	Execucomp
Indcompcom	An indicator variable that equals one if the compensation committee is comprised wholly of independent directors.	Execucomp
Market competition variables		
CP _oriented	The ratio of advertising expenses to sales revenue.	Compustat
Fluidity	A degree of change in rival firm's products relative to a firm's products.	Hoberg et al. (2016)
Similarity	The degree to which a firm's product perspective is similar to that of rivals.	Hoberg and Phillips (2010, 2014)
Earning announcement, crash risk, and information asymmetry		
CAR[-1;1]	Cumulative abnormal returns in event window [-1; 1] of the earnings announcements. The estimation window is [-230; −31] and requires at least 100 observations in the estimation window. Returns in event windows are adjusted by the Fama-French 3-factor model.	CRSP
Coverage	Number of estimates that constitute EPS Forecast from I/B/E/S Unadjusted Summary Data (WRDS file name is *ibes.statsumu*).	I/B/E/S
Firm age	Firm age is calculated from the first year with non-missing price in the CRSP database	CRSP
SUE_mean	Difference between actual EPS and the mean forecasted EPS, scaled by the standard deviation in the forecasted EPS by analysts.	I/B/E/S
SUE_median	Difference between actual EPS and the median forecasted EPS, scaled by the standard deviation in the forecasted EPS by analysts.	I/B/E/S
Other measures of CEO compensations		
Delta	The CEO's change in wealth (expressed in thousands of dollars) for a 1% increase in the firm's stock price.	Execucomp
Firm related wealth	The sum of the value of the stock and option portfolio held by the CEO (Daniel et al., 2013).	Execucomp
Non-equity-based	The total compensation (TDC) minus equity-based compensation (Chhaochharia & Grinstein, 2009).	Execucomp
Vega	The CEO's change in wealth (expressed in thousands of dollars) for a 1% increase in the annualized standard deviation of the firm's stock returns.	Execucomp

CRSP: Center for Research in Security Prices.

Appendix B. Sample distribution by SIC classification

SIC2	Observations	With CEO Twitter	Without CEO Twitter	Standard Industrial Classification

(*continued on next page*)

(*continued*)

SIC2	Observations	With CEO Twitter	Without CEO Twitter	Standard Industrial Classification
01	12	0	12	Agricultural Production
02	13	0	7	Agricultural Production livestock and animal
07	9	0	9	Agricultural services
10	31	0	31	Metal, Mining
12	16	3	13	Coal Mining
13	475	19	456	Oil & Gas Extraction
14	65	0	65	Nonmetallic Minerals, Except Fuels
15	141	13	128	General Building Contractors
16	79	0	79	Heavy Construction, Except Building
17	58	0	58	Special Trade Contractors
20	470	54	416	Food & Kindred Products
21	38	6	32	Tobacco Products
22	45	86	45	Textile Mill Products
23	164	0	164	Apparel & Other Textile Products
24	77	0	77	Lumber & Wood Products
25	104	0	104	Furniture & Fixtures
26	156	0	156	Paper & Allied Products
27	106	0	106	Printing & Publishing
28	1,086	86	1,000	Chemical & Allied Products
29	102	25	77	Petroleum & Coal Products
30	127	26	101	Rubber & Miscellaneous Plastics Products
31	67	0	67	Leather & Leather Products
32	50	0	50	Stone, Clay, & Glass Products
33	250	15	235	Primary Metal Industries
34	252	7	245	Fabricated Metal Products
35	809	165	734	Industrial Machinery & Equipment
36	1,094	155	936	Electronic & Other Electric Equipment
37	463	26	437	Transportation Equipment
38	942	152	790	Instruments & Related Products
39	128	23	105	Miscellaneous Manufacturing Industries
40	51	0	51	Railroad Transportation
41	1	0	1	Local & Interurban Passenger Transit
42	142	13	129	Trucking & Warehousing
44	61	0	61	Water Transportation
45	124	34	90	Transportation by Air
47	68	32	36	Transportation Services
48	250	65	185	Communications
49	795	47	748	Electric, Gas, & Sanitary Services
50	354	57	297	Wholesale Trade – Durable Goods
51	219	24	195	Wholesale Trade – Nondurable Goods
52	50	13	37	Building Materials & Gardening Supplies
53	144	9	135	General Merchandise Stores
54	56	15	41	Food Stores
55	135	14	121	Automotive Dealers & Service Stations
56	278	26	252	Apparel & Accessory Stores
57	62	17	45	Furniture & Home furnishings Stores
58	283	93	190	Eating & Drinking Places
59	241	59	182	Miscellaneous Retail
60	1,149	106	1,043	Depository Institutions
61	132	16	116	Non depository Institutions
62	385	49	336	Security & Commodity Brokers
63	679	39	640	Insurance Carriers
64	100	13	87	Insurance Agents, Brokers, & Service
65	42	8	34	Real Estate
67	1,030	40	990	Holding & Other Investment Offices
70	37	0	37	Hotels & Other Lodging Places
72	62	0	62	Personal Services
73	1,498	519	979	Business Services
75	33	0	33	Auto Repair, Services, & Parking
78	44	26	18	Motion Pictures
79	86	13	73	Amusement & Recreation Services
80	280	55	225	Health Services
82	58	21	37	Educational Services
83	0	0	0	Social Services
87	263	48	215	Engineering & Management Services
99	56	24	32	Non-Classifiable Establishments

Appendix C. Example of CEOs who are followed by other CEOs.This table shows example of CEOs who are followed by other CEOs on Twitter

Followers			Followings		
Executive name	Twitter account	Company name	Executive name	Twitter account	Company name
Dennis A. Muilenburg	BoeingCEO	Boeing	Guillaume Faury Tory Bruno	GuillaumeFaury torybruno	Airbus United Launch Alliance
Guillaume Faury	GuillaumeFaury	Airbus	Dennis A. Muilenburg Tory Bruno	BoeingCEO torybruno	Boeing United Launch Alliance
Tim Cook	Tim Cook	Apple	Sundar Pichai Satya Nadella	sundarpichai satyanadella	Google Microsoft
Marissa A. Mayer	marissamayer	Yahoo	Sundar Pichai Jeremy Stoppelman	sundarpichai jeremys	Google Yelp
Sundar Pichai	sundarpichai	Google	Marissa A. Mayer Jeff Bezos Satya Nadella Tim Cook Susan Wojcicki	marissamayer JeffBezos satyanadella tim_cook Susan Wojcicki	Yahoo Amazon Microsoft Apple Youtube

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